

KANIKA SHARMA

**Research | Biomedical Science
| Biotechnology | Writing**

India | kanika06123@gmail.com

PROFILE

I am a research enthusiast, an aspiring biomedical scientist, passionate about biology, biotechnology, quantum mechanics and overall STEM. Currently in highschool final year pursuing, Physics, Chemistry, Biology along with Computer Science . While actively participating in biomedical science and biotechnology courses. Founder of STEM based community which works to provide an idea or solution for real world problems in STEM. Aiming to evolve in the field of STEM.

EDUCATION

Delhi Public School

12th Grade

Core Subjects: Physics, Chemistry, Biology
and Computer Science

CERTIFICATIONS

- STEM Research Scholar
- Business Analysis & Process Management project
- Certificate of achievement in UX Design with company project at Infinity cars
- Certificate of achievement in Data and Business Analysis at Samsonite
- Certificate of young programmer in python
- Certificate of completing mini MBA course

SKILLS

- Content Writing
 - Scientific Writing
 - Copy writing
 - Biology
 - Biomedical Research
 - Research
 - Team management
 - Entrepreneurship
 - Web Design
 - Data Analysis
 - Business Analysis & Strategy
 - UX Designing
 - Python
 - SQL
 - HTML / CSS
 - Medical Research
 - Leadership
 - Blogging
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RESEARCH EXPERIENCE

THE NEW YORK ACADEMY OF SCIENCES

Research Topic : Solar Technology and it's efficiency

October 2025 - January 2026

- Working on research to make retrofit the current solar technology.
- Aiming to make the existing solar technology and power energy more efficient and viable across large scaled areas.
- Leading the team and designing solutions, while focusing on the present demerits of the technology

Research Topic : Ethical AI

September 2024 - December 2024

MENTORED BY : ALYSSA AFARI, MOUNT ST MARY'S UNIVERSITY

- Led the team to come up with a logical solution for making a safer use of Artificial Intelligence.
- Worked on team research and real life case studies associated and relevant with it.
- Developed skills in research, artificial intelligence, team leading, team building and team management.
- Developed a basic website to spread the awareness.

Research Topic : Living in the Extremes

February 2025 - April 2025

- Worked on solutions to survive in the areas with one extreme weather.
- Did research on certain geographical areas around the globe.
- Led the team for research, case studies, surveys etc.
- Worked on a team research paper to survive accordingly.

Topic : Human Centred AI

March 2026 - April 2026

- Worked in drug discovery research
- Lead team to work on drug discovery involving Artificial Intelligence
- Created Hypothesis
- Working with research on artificial intelligence, which helps human scientists effeciently.

ROARING CUBS COLLECTIVE ~ Scientific Review

Research Topic : Bluelight

MENTORED BY : MALVINA MILLET, HARVARD MEDICAL SCHOOL

October 2024 - January 2025

- Conducted research on Bluelight exposure focusing on it's effect on human vision, health and circadian rhythm, focusing on the correlation between screen time and retinal health.
- Applied APA-format literature review methods and used scientific databases such as PubMed, Google Scholars and Science Direct to collect and analyze data.
- Independent literature based-analytical research while mentored by Ms. Malvina Millet.
- Developed critical skills in scientific writing, data visualization and organization

Research Topic : Advancing Alzheimer's Therapy: Nanomedicine, Cell-Specific Drug Discovery, and Anti-Amyloid Strategies

MENTORED BY : CHRISTOPHER ARELLANO REYES, UNIVERSITY OF SOUTHERN CALIFORNIA

October 2025- January 2026

- Conducted an in-depth scientific review on Alzheimer's disease, focusing on nanomedicine, blood-brain barrier (BBB) repair, cell-type-specific drug discovery, and anti-amyloid therapeutic strategies.
- Analyzed Alzheimer's pathophysiology, including amyloid- β plaque formation, tau protein aggregation, neuroinflammation, and neuronal degeneration.
- Explored advanced nanotechnology-based interventions targeting BBB dysfunction and evaluated their role in improving drug delivery and cognitive outcomes.
- Investigated cell-type-specific therapeutic approaches using transcriptomics and drug perturbation databases to identify targeted treatment strategies.
- Reviewed and compared emerging anti-amyloid therapies, including monoclonal antibodies and enzyme-targeting approaches, assessing their efficacy, limitations, and clinical impact.
- Examined clinical trial data (e.g., Lecanemab and Donanemab) to evaluate real-world therapeutic outcomes and disease progression reduction.
- Studied novel drug repurposing approaches and multi-target treatment strategies integrating computational biology and clinical evidence.
- Applied literature review methodologies using peer-reviewed scientific sources to synthesize and critically evaluate current Alzheimer's research.
- Developed strong skills in scientific writing, critical analysis, and interpretation of biomedical data and clinical studies.
- Conducted independent, mentor-guided research under Christopher Arellano Reyes, focusing on translating theoretical insights into future therapeutic directions.

PROJECT EXPERIENCE

SAMSONITE

Data Analysis Company Project & Business Analysis and Strategy

January 2022 - February 2022

- Worked on analyzing data along with a mentor and a diverse team of students.
- Analyzed data for Samsonite, to get better understanding of market.
- Analyzed data for Samsonite, working on it's expansion in India.
- Worked on its strategies, marketing and sales.
- Used platforms like google studios, to work and deal with data.
- Developed skills in analyzing data, team communication, statistics, marketing, market research, SEOs, google studio, team collaboration.

INFINITY CARS

UX Deisgning

March 2022

- Worked on prototyping Infinity Car's website.
- Learned strategies and psychological techniques on how user experiences work.
- Worked on peer reviewed usability test.
- Used platforms like marvel, to design prototypes.
- Developed skills in wireframing, user interface, user experience, color palette, understanding users.